

Artificial Intelligence



Introduction to AI & Generative AI

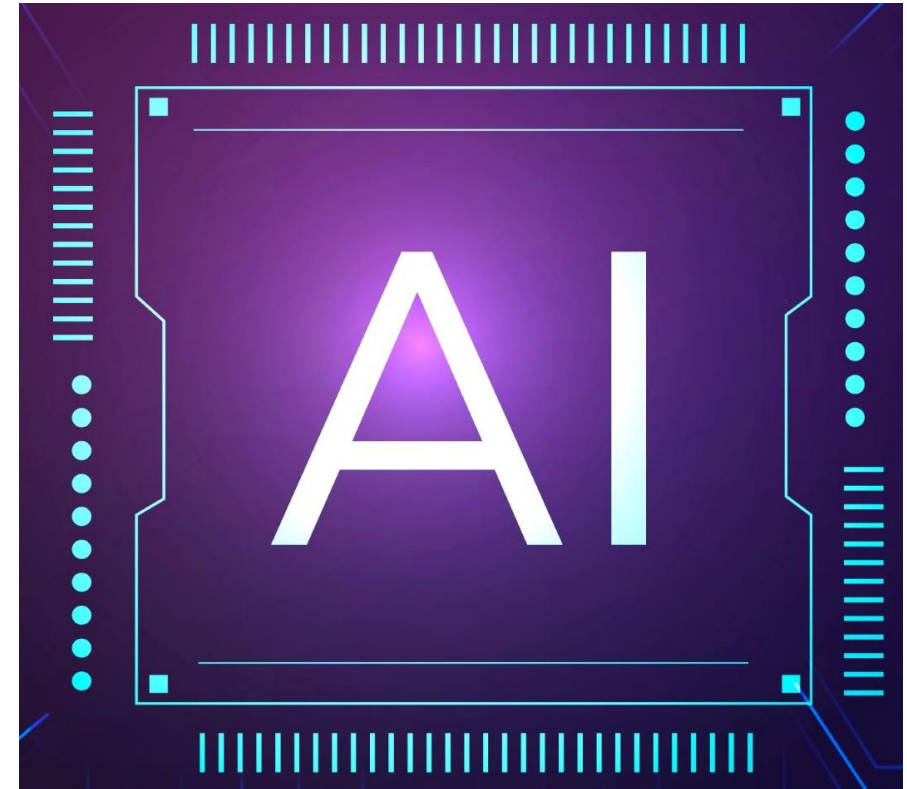
- Introduction to AI
- Neural Networks
- Natural Language Processing
- Computer Vision

Labs

- Students to use ChatGPT for various prompts
- Asking the AI Models to create a phishing email

Introduction to AI

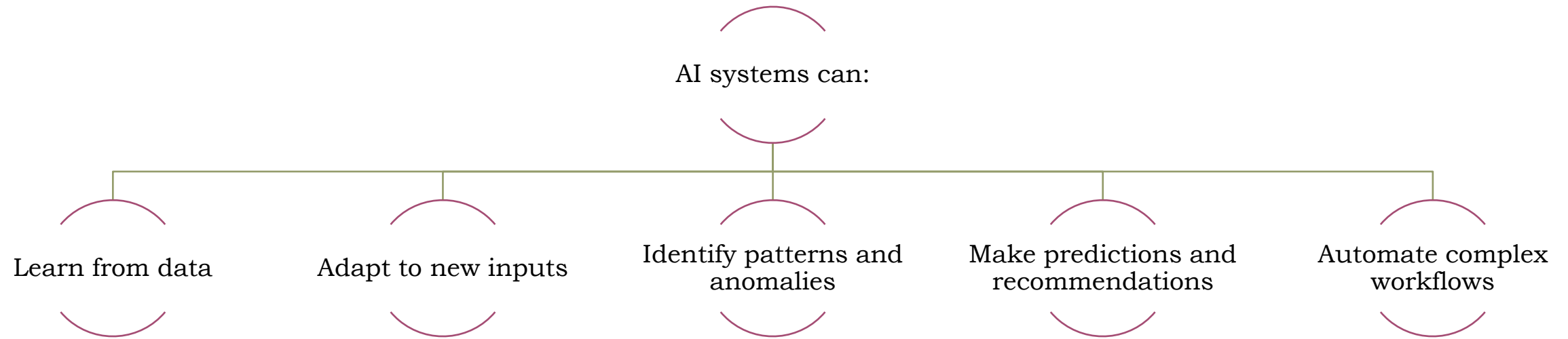
- Artificial Intelligence (AI) is a branch of computer science focused on building systems that can simulate human intelligence.
- These systems are designed to learn, reason, perceive, understand language, and make decisions with minimal human intervention.
- AI enables machines to think, learn, and act intelligently.



Why AI Is Important?

- AI is transforming how organizations:
 - Analyze data
 - Automate processes
 - Improve decision-making
 - Enhance customer experience
 - Detect threats and anomalies
- AI enables systems to perform tasks that would otherwise require human intelligence—faster, more consistently, and at scale.

Key Characteristics of AI



Types of Artificial Intelligence

Narrow AI (Weak AI)

- Designed for specific tasks
- Examples: spam filters, face recognition, chatbots, recommendation systems
- This is the only type of AI in practical use today.

General AI (Strong AI)

- Human-level intelligence across multiple domains
- Can reason and learn like a human
- Currently theoretical; not yet achieved.

Super AI

- Intelligence exceeding human capabilities
- Hypothetical and future-focused.



Core AI Domains

Machine Learning (ML)

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Deep Learning

- A subset of ML using neural networks with multiple layers.
- Used in image recognition, speech processing, NLP

Natural Language Processing (NLP)

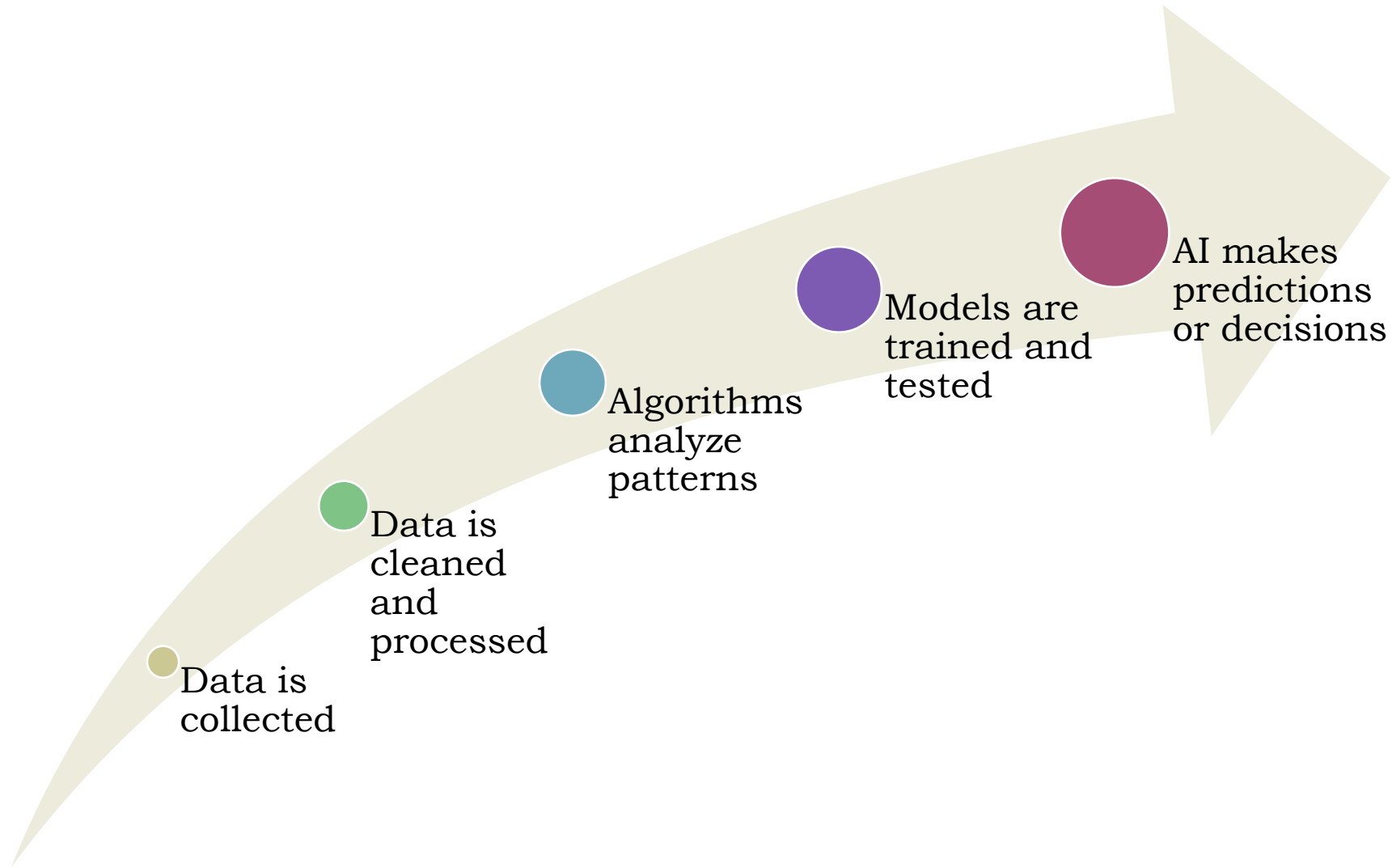
- Chatbots
- Voice assistants
- Language translation

Computer Vision

- Allows machines to interpret visual data.
- Facial recognition
- Object detection
- Medical imaging

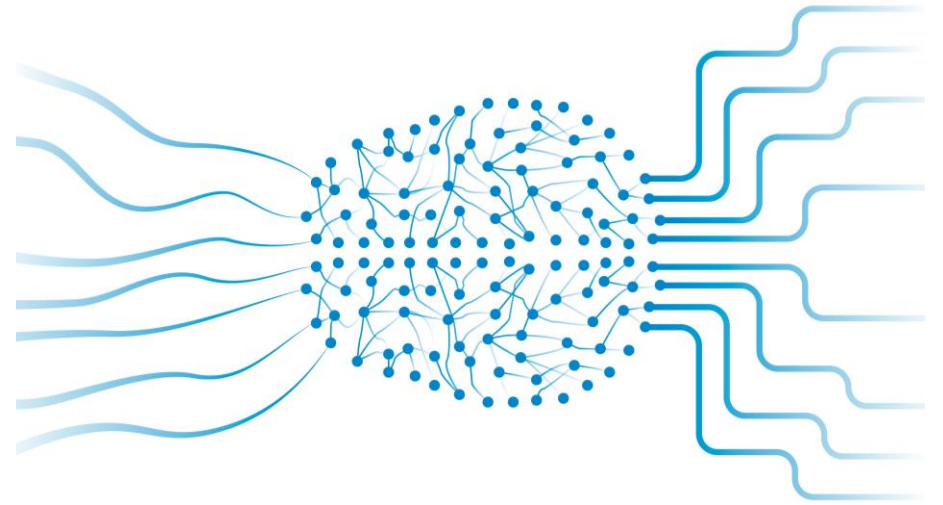


How AI Works



Neural Networks

- A Neural Network is a computational model inspired by the human brain.
- It consists of interconnected processing units called neurons that work together to recognize patterns, learn from data, and make predictions or decisions.
- Neural networks are a core building block of *Artificial Intelligence (AI) and Machine Learning (ML)*, especially Deep Learning.
- A neural network is a machine learning model composed of interconnected neurons arranged in layers, capable of learning complex patterns from data through training.



Why Neural Networks Are Important

- Neural networks excel at:
 - Pattern recognition
 - Image and speech processing
 - Natural language understanding
 - Complex decision-making
 - Learning from large datasets
- They can model non-linear relationships that traditional algorithms struggle with.

Basic Structure of a Neural Network

- **Input Layer**

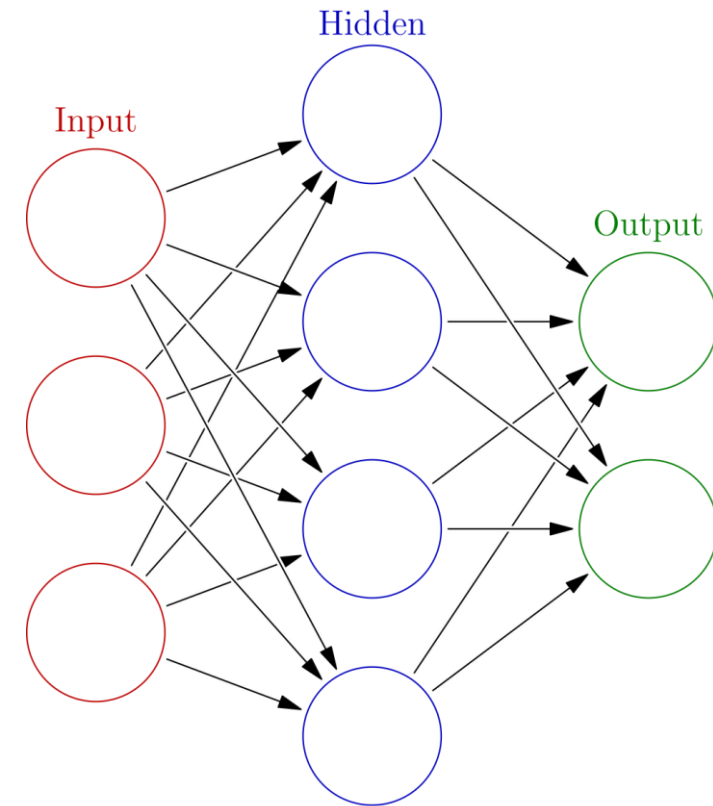
- Receives raw data
- Example: pixels of an image, numerical features, text embeddings

- **Hidden Layers**

- Perform computations and feature extraction
- May consist of one or many layers
- The more hidden layers, the “deeper” the network

- **Output Layer**

- Produces final prediction or classification
- Example: spam vs non-spam, price prediction, object label



How a Neuron Works?

Each neuron performs three main steps:

1. Weighted Sum

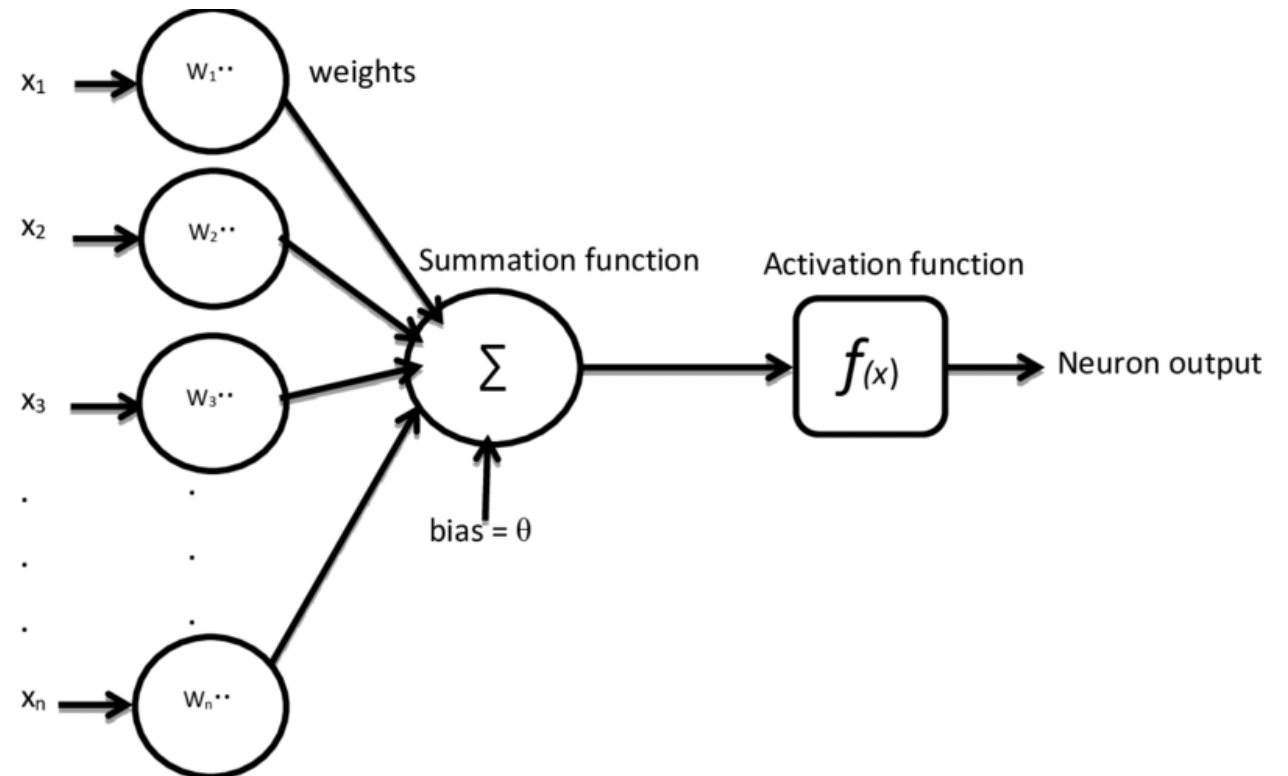
- Inputs are multiplied by weights and summed.

2. Add Bias

- A bias term shifts the result.

3. Activation Function

- Applies a non-linear transformation.



Mathematically:

$$\text{Output} = \text{Activation}(\Sigma (\text{Input} \times \text{Weight}) + \text{Bias})$$

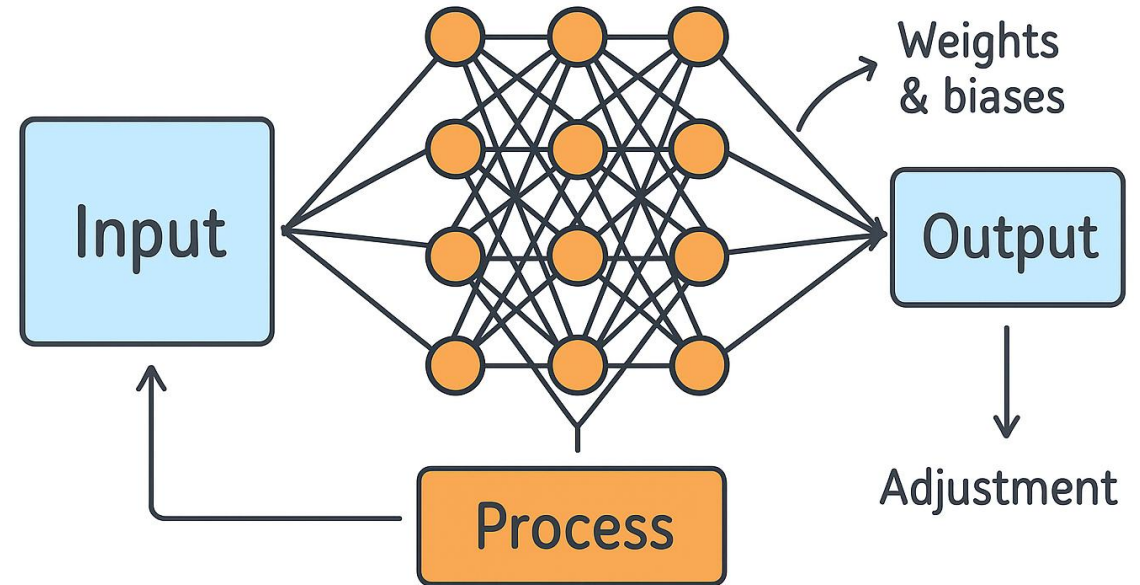
How Neural Networks Learn?

■ Training Phase

- Network makes predictions
- Error is calculated using a loss function
- Error is propagated backward (backpropagation)
- Weights are updated using gradient descent
- This process repeats until error is minimized.

■ Inference Phase

- Trained model makes predictions on new data
- No weight updates occur

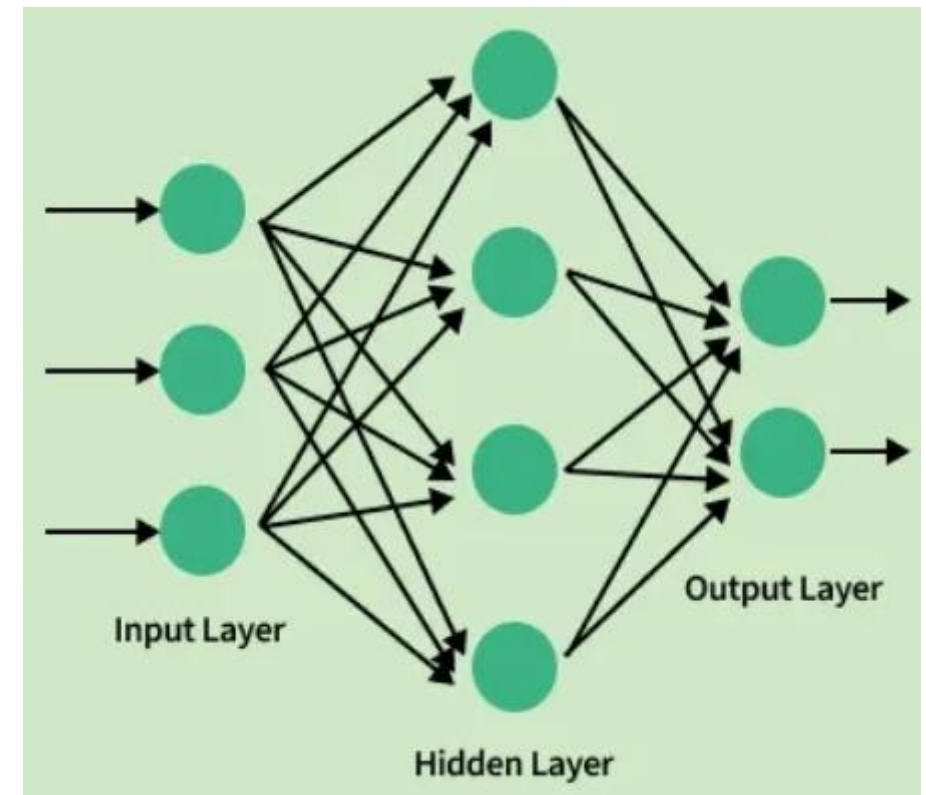


Types of Neural Networks

- Feedforward Neural Network (FNN)
- Convolutional Neural Network (CNN)
- Recurrent Neural Network (RNN)
- Long Short-Term Memory (LSTM)
- Transformer Networks

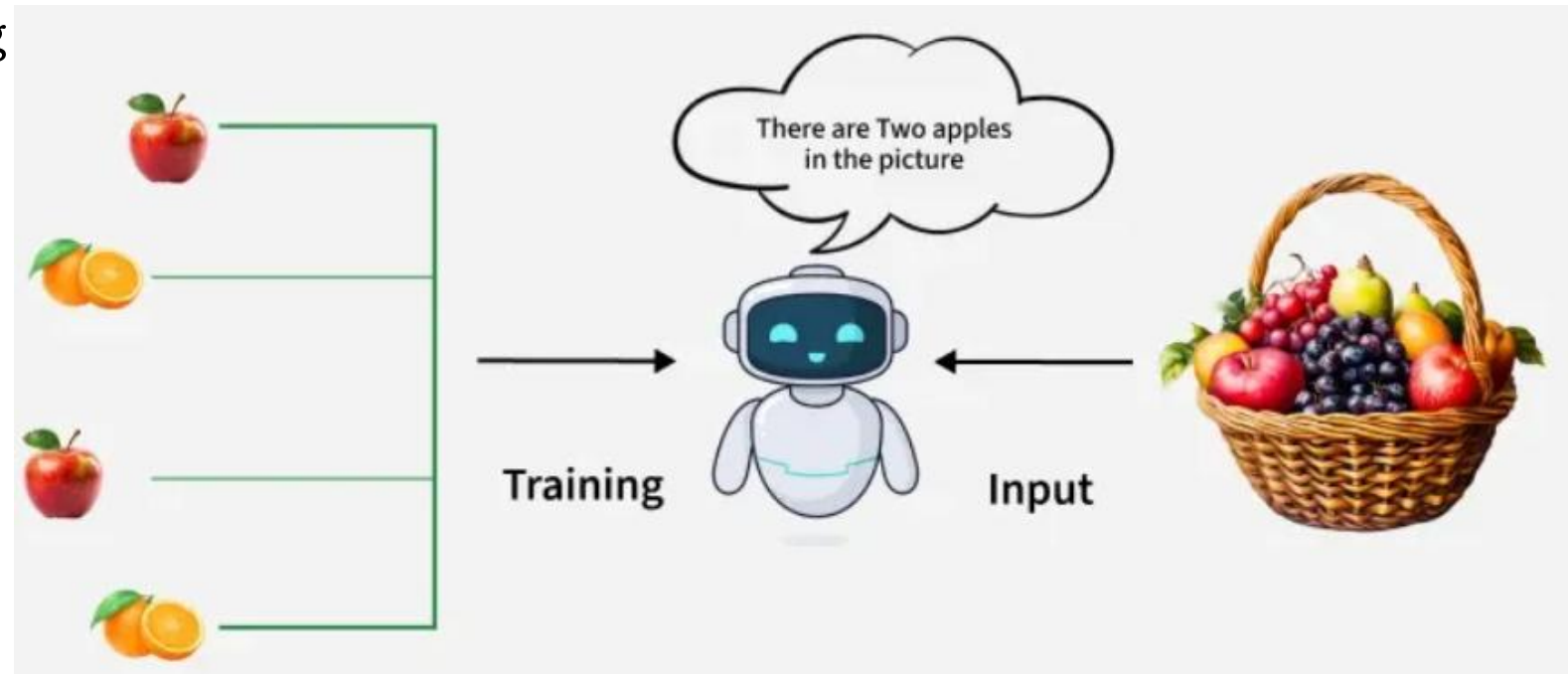
Feedforward Neural Network (FNN)

- Data flows in one direction i.e, from input to output
- Used for basic classification and regression
- No loops or feedbacks.
- Application of FNN are:
 - Image Recognition
 - Speech Recognition
 - Credit Scoring
 - Predictive Analysis
 - Medical Diagnosis
 - Anomaly detection.



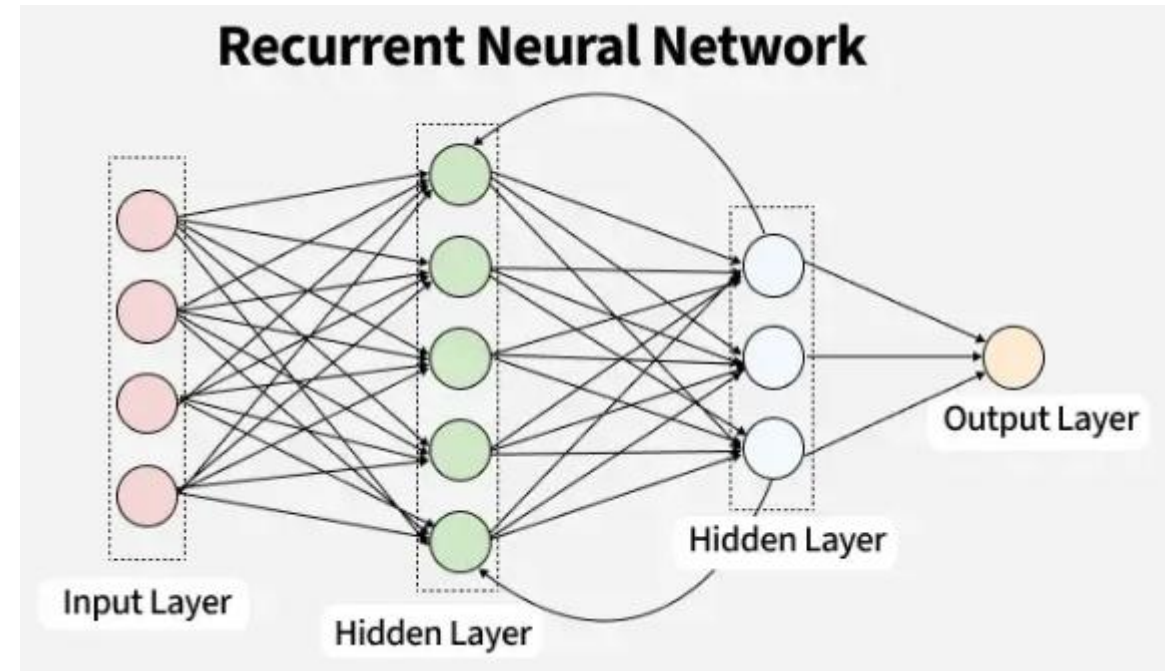
Convolutional Neural Network (CNN)

- Specialized for image and video data
- Uses convolution and pooling
- Applications of CNN
 - Automotive
 - Social Media
 - Health Care
 - Law Enforcement
 - Optical character recognition (OCR)



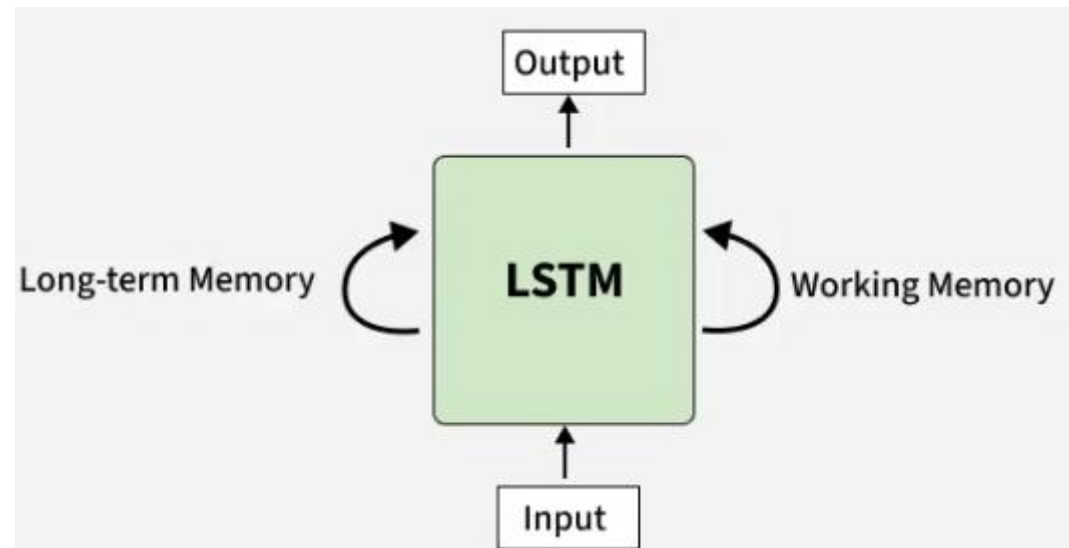
Recurrent Neural Network (RNN)

- Designed for sequential data
- Maintains memory of previous inputs
- It feed information back into the network at each step.
- Application of RNN:
 - Time-Series Prediction
 - Natural Language Processing (NLP)
 - Speech Recognition
 - Image and Video Processing



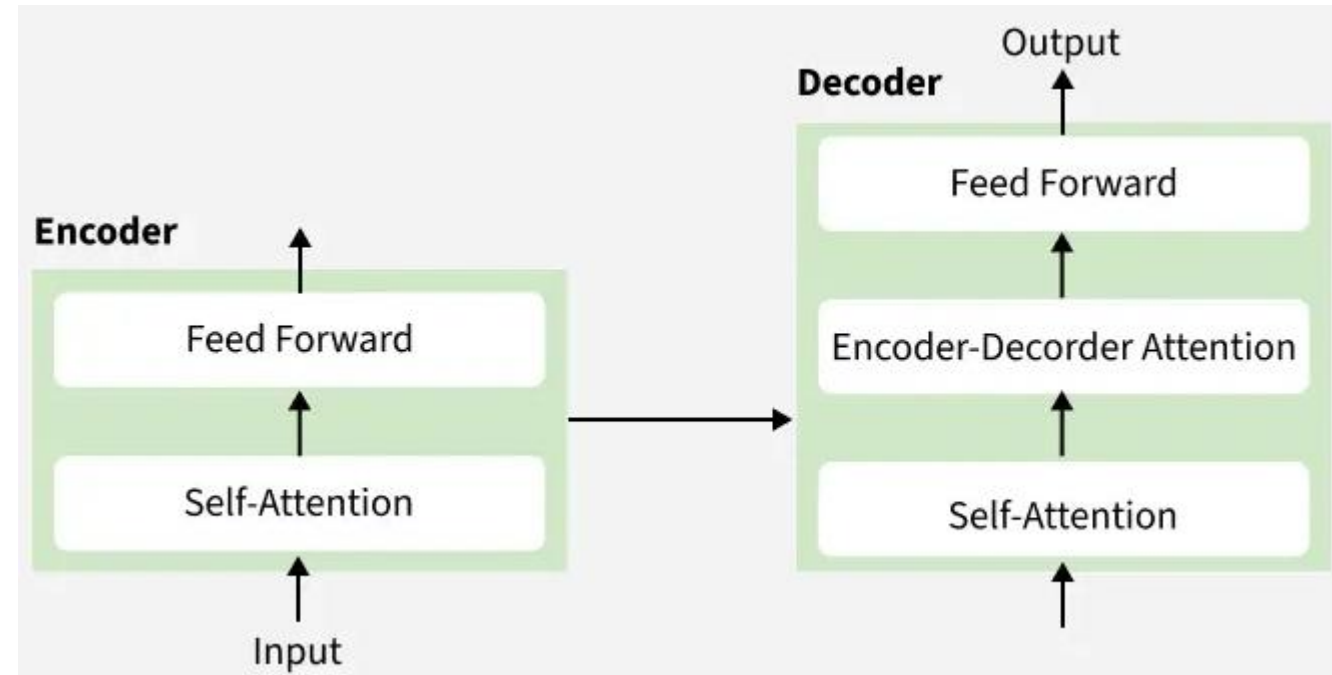
Long Short-Term Memory (LSTM)

- Advanced RNN for long-term dependencies
- Used in speech and language models
- LSTMs can capture long-term dependencies in sequential data making them ideal for tasks like language translation, speech recognition and time series forecasting.
- Applications of LSTM:
 - Image Captioning
 - Machine Translation
 - Music Generation
 - Auto Handwriting Generation



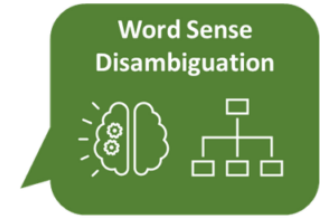
Transformer Networks

- Uses attention mechanisms
- Powers modern NLP models like GPT and BERT
- Provides better performance than any of the other models due to their capability to process natural language sufficiently and handle tasks related to machine translation, generating text and document summarization.



Natural Language Processing (NLP)

- Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables computers to understand, interpret, generate, and interact with human language in both text and speech form.
- NLP bridges the gap between:
 - Human language (unstructured, ambiguous)
 - Machine understanding (structured, logical)
- NLP allows machines to:
 - Read documents
 - Understand user intent
 - Extract insights from text
 - Communicate naturally with humans



Natural Language Processing



Key Objectives of NLP

NLP systems aim to:

- Understand language (Natural Language Understanding – NLU)
- Generate language (Natural Language Generation – NLG)
- Translate language
- Extract meaning and entities
- Analyse sentiment and intent



Core Components of NLP

Natural Language Understanding (NLU)

- Focuses on interpreting user input:
 - Intent detection
 - Entity recognition
 - Context understanding

Natural Language Generation (NLG)

- Focuses on producing human-like responses:
 - Text summarization
 - Chatbot responses
 - Automated reporting



NLP Processing Pipeline

- **Step 1: Text Preprocessing**

- Tokenization (splitting text into words)
- Lowercasing
- Stop-word removal
- Stemming and lemmatization

- **Step 2: Feature Extraction**

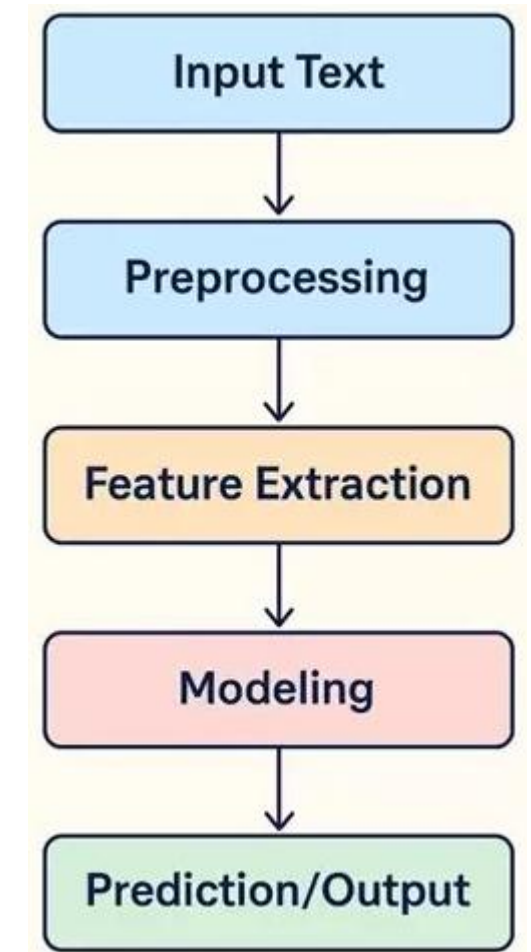
- Converts text into numerical form:
- Bag of Words (BoW)
- TF-IDF
- Word embeddings (Word2Vec, GloVe)
- Contextual embeddings (BERT, GPT)

- **Step 3: Modelling**

- Machine learning models
- Deep learning models
- Transformer-based architectures

- **Step 4: Evaluation**

- Accuracy
- Precision, Recall, F1-score
- BLEU score (translation)
- ROUGE score (summarization)



NLP Models

Rule-Based Models

- Predefined linguistic rules
- Limited scalability

Statistical Models

- Probabilistic approaches
- N-grams, HMMs

Machine Learning Models

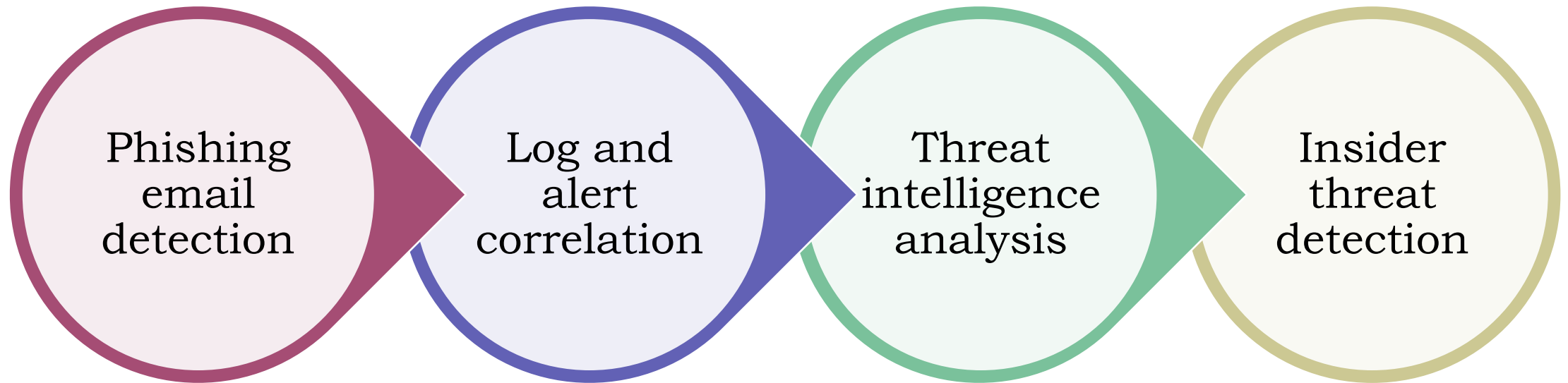
- Naive Bayes
- SVM
- Logistic Regression

Deep Learning Models

- RNN, LSTM
- CNN for text
- Transformers (BERT, GPT, T5)



NLP in Cybersecurity



Benefits of NLP



Computer Vision

- Computer Vision (CV) is a branch of Artificial Intelligence (AI) that enables machines to interpret, analyze, and understand visual data such as images and videos in a way similar to human vision.
- The objective of computer vision is to extract meaningful information from pixels and use it for decision-making.
- Visual data constitutes the majority of digital information today:
 - Images, Videos
 - CCTV footage, Medical scans
- Computer vision automates visual understanding at scale, speed, and accuracy that humans cannot achieve manually.



How Computer Vision Works?

Image or video capture



Preprocessing (resize, normalize, noise reduction)



Feature extraction



Model inference (prediction/classification)



Post-processing and decision-making

Core Components of Computer Vision

▪ Image Acquisition

- Cameras
- Sensors
- Drones
- Medical imaging devices

▪ Feature Extraction

- Identifies important patterns:
 - Edges
 - Shapes
 - Textures
 - Colours

▪ Model Training

- Uses labeled or unlabeled data to train:
 - CNNs
 - Vision Transformers (ViT)

▪ Inference and Interpretation

- Models predict:
 - Object category
 - Location
 - Movement
 - Anomalies

▪ Image Preprocessing

- Improves image quality and consistency:
 - Grayscale conversion
 - Noise reduction
 - Contrast enhancement
 - Normalization

Core Components of Computer Vision

- **Model Training**

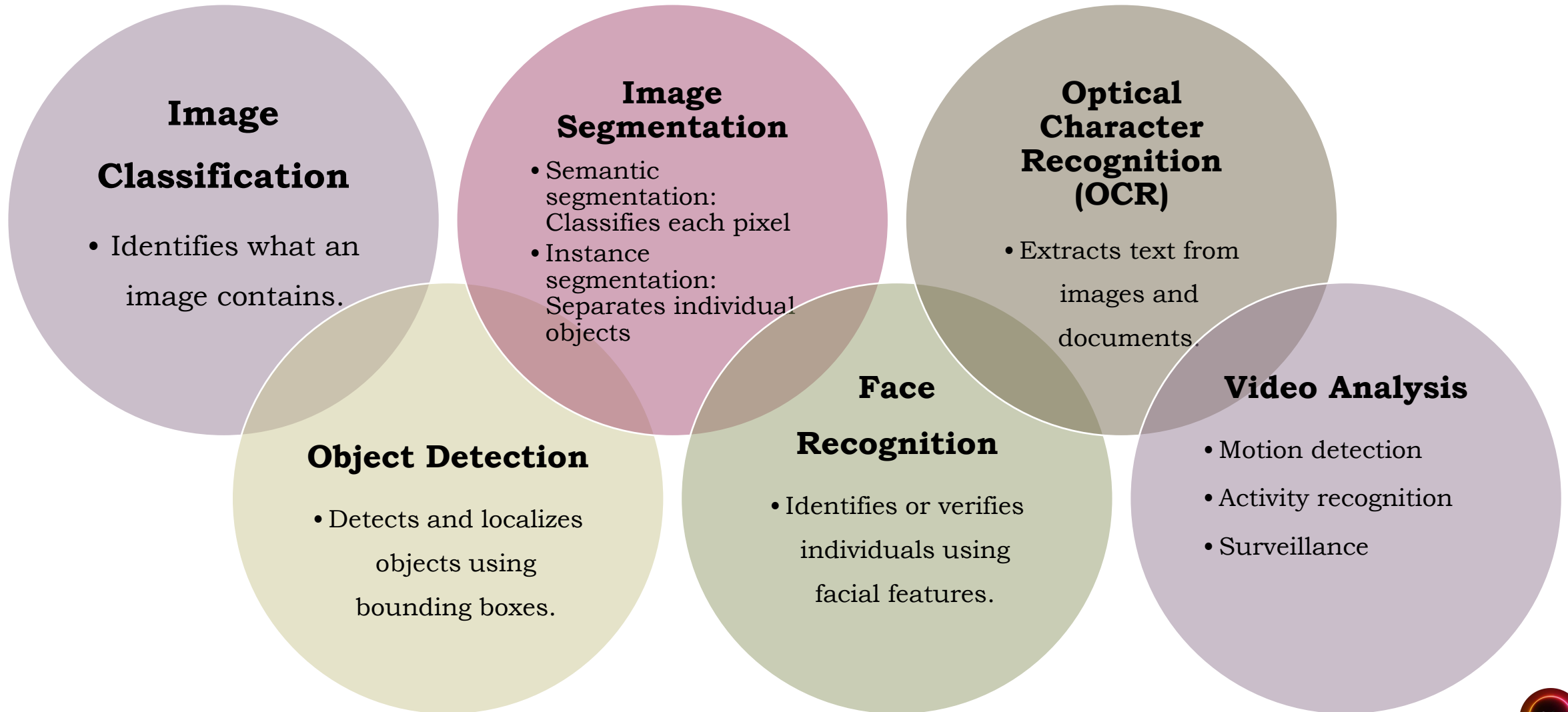
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- **Inference and Interpretation**

- Models predict:
 - Object category
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Key Computer Vision Tasks



Computer Vision vs Human Vision

Aspect	Human Vision	Computer Vision
Speed	Limited	High
Accuracy	Variable	Consistent
Fatigue	Yes	No
Learning	Experience	Data-driven

Benefits of Computer Vision



Automation of visual tasks

High accuracy

Real-time processing

Scalability

Generative AI (GenAI)



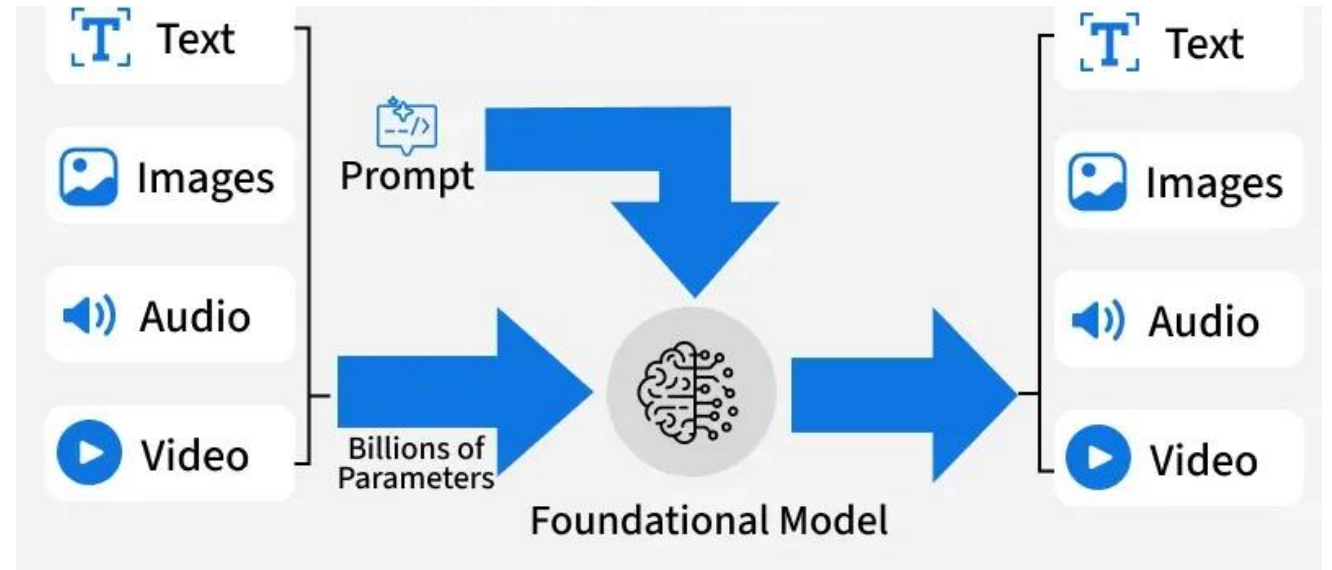
Generative AI

- **Generative AI**

- Supervised vs Unsupervised Learning
- Discriminative AI vs. Generative AI
- Brief History of Generative AI
- A Peek into Generative AI Models
- Large Language Models(LLMs)
- Introducing ChatGPT – Hands-on How does Model understand Text
- How does Model understand Text

What is Generative AI?

- Generative AI is an AI technology that learns from large datasets to generate new, realistic, and contextually relevant content in various formats such as text, images, audio, video, and code.
- It is a category of AI systems designed to create new, original content like text, images, audio, video, code or synthetic data by learning patterns and structures from existing data.
- Generative AI produces new outputs that resembles human-created content.



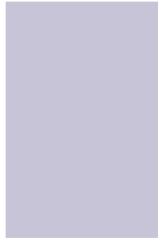
Key Characteristics of Generative AI



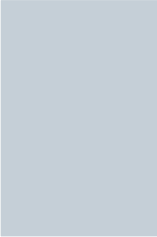
Content Generation



Learning from Data Patterns



Contextual Understanding



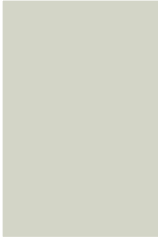
Adaptability/Versatility



Automation of Creative Tasks

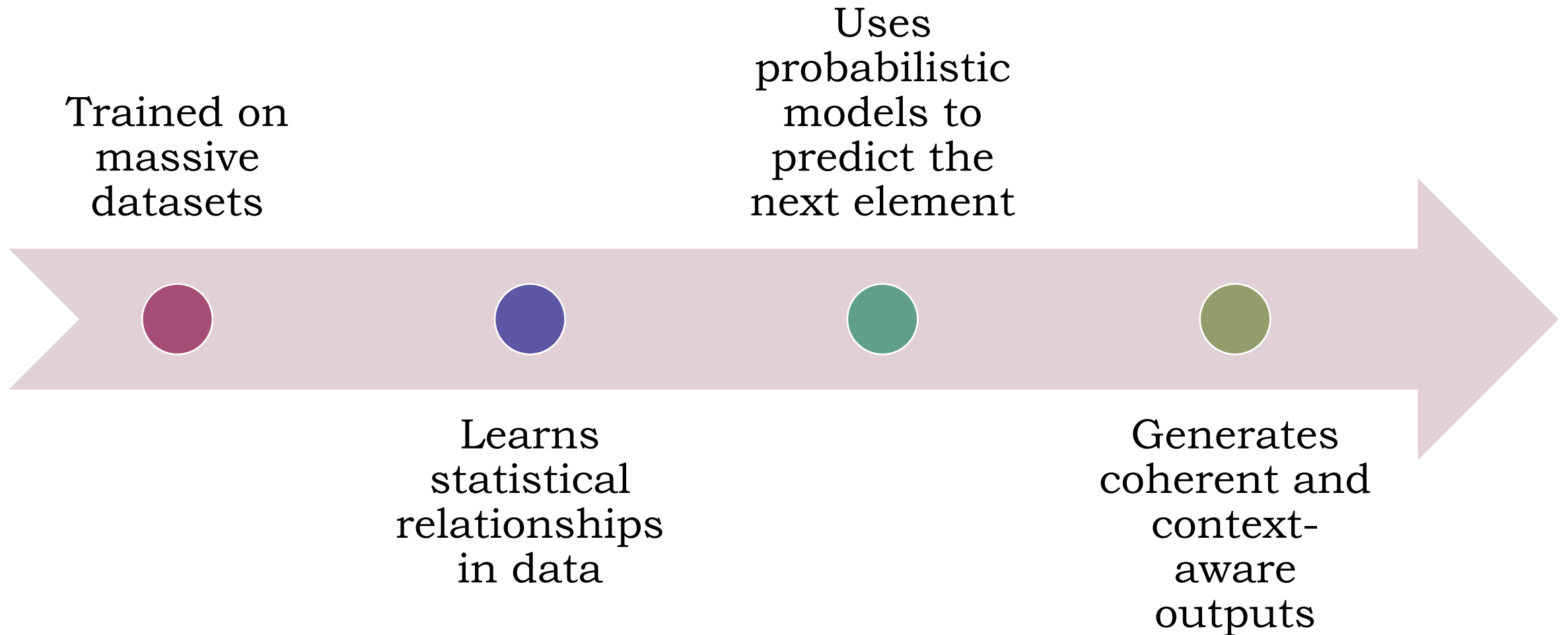


Scalability and Speed

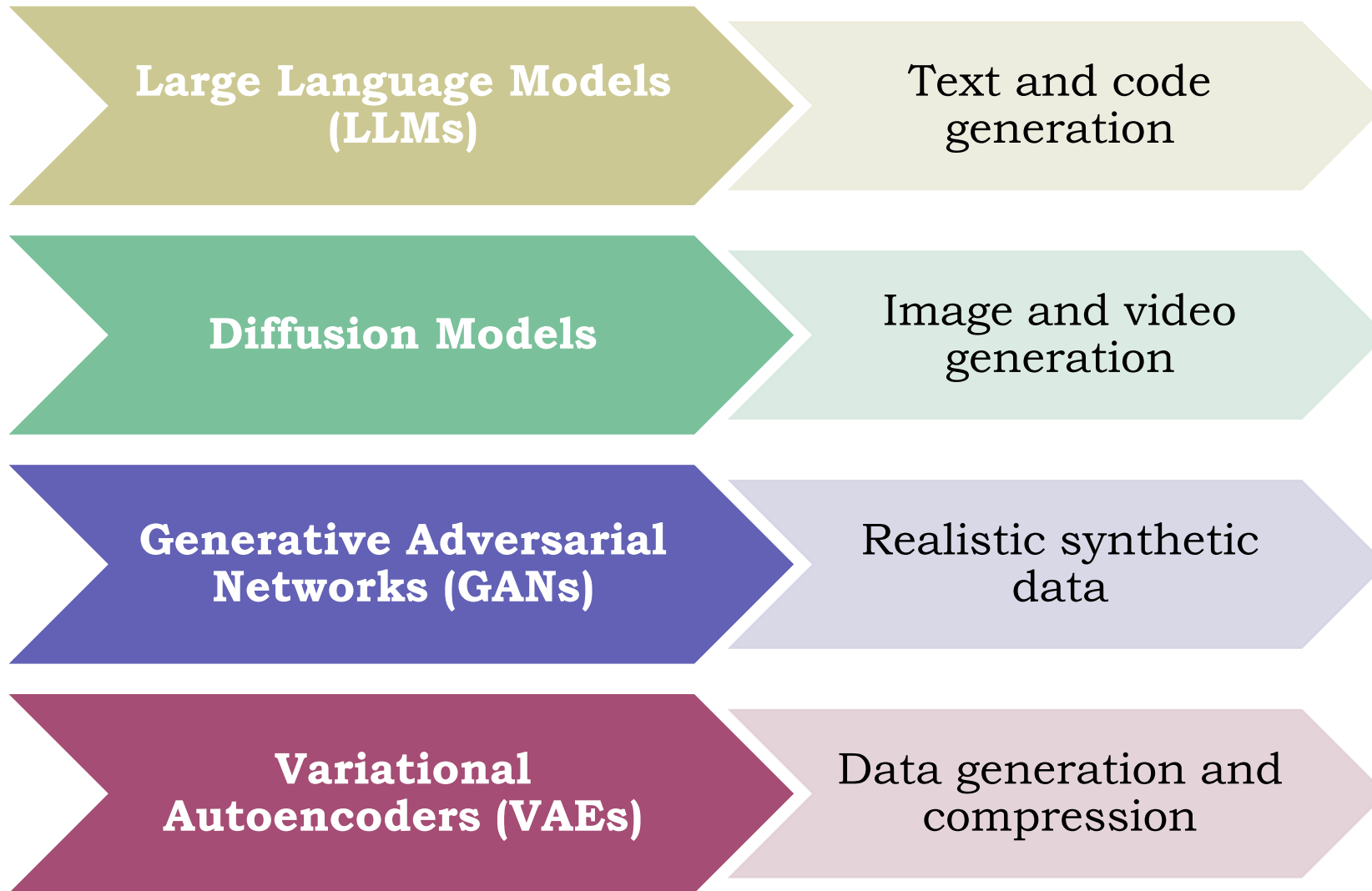


Emergent Abilities

How Generative AI Works



Common Generative AI Models



Applications of Generative AI

▪ Text

- Chat bots
- Virtual assistants
- Content creation

▪ Images

- Digital arts,
- Product design
- Fashion

▪ Audio & Speech

- Enables natural-sounding
- Music composition
- Personal voice cloning

▪ Video

- Animation, special effects
- Gaming, marketing videos
- Realistic training simulations.

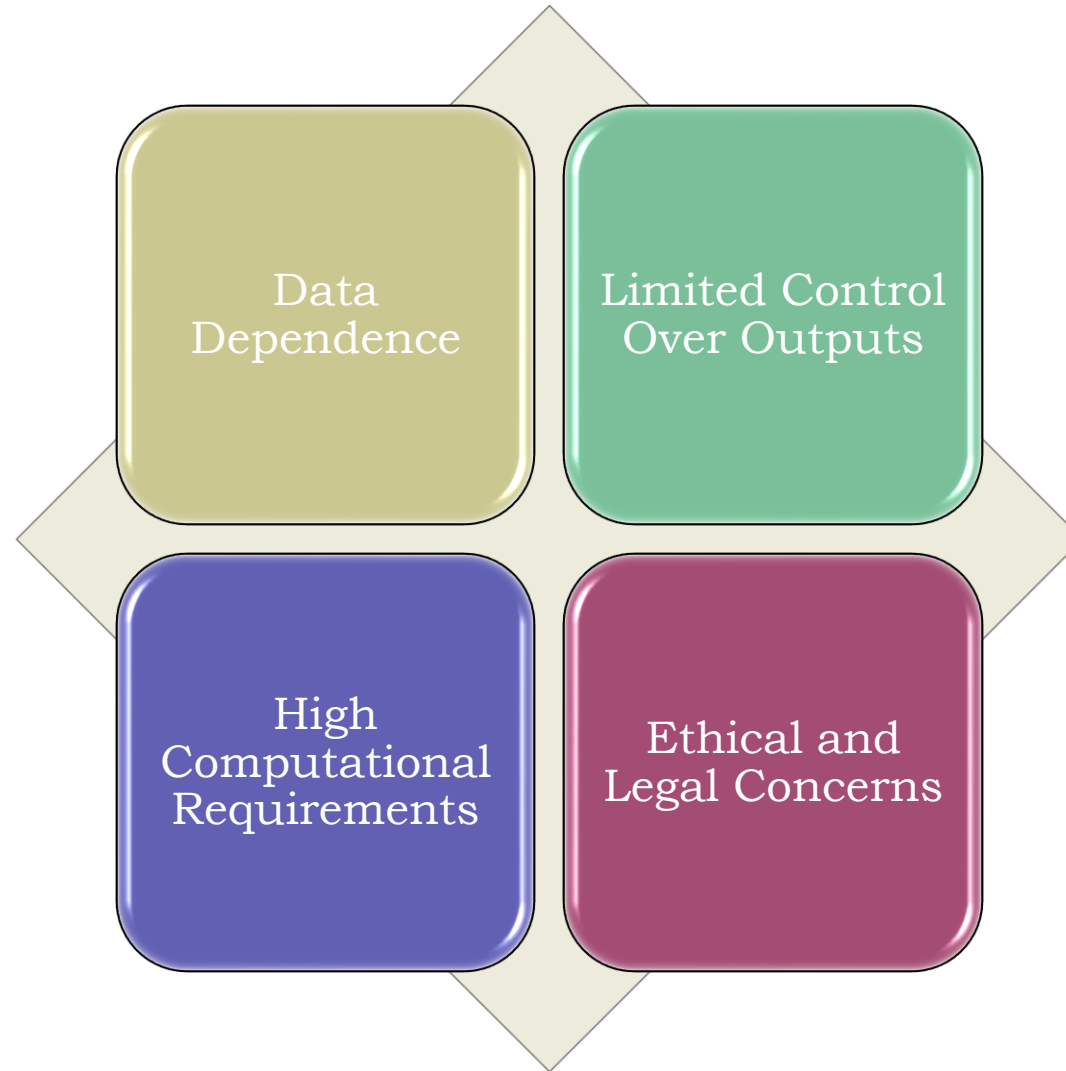
▪ Business Use case

- Enhances customer support with AI agents.
- Boots knowledge discovery using RAG systems.
- Accelerates drug discovery.
- Improves data-driven decision-making.

Advantages of GenAI



Disadvantages of GenAI



Examples of Generative AI

- **Human-like text generation**

- Generative AI models, including the GPT series, can create articles, stories, and social media posts, enhancing content creation and marketing efforts

- **Image creation**

- Models such as DALL-E can generate unique images from text prompts, allowing users to produce artwork or designs that reflect specific concepts or styles—perfect for art direction, product design, and visual design in gaming.

Examples of Generative AI

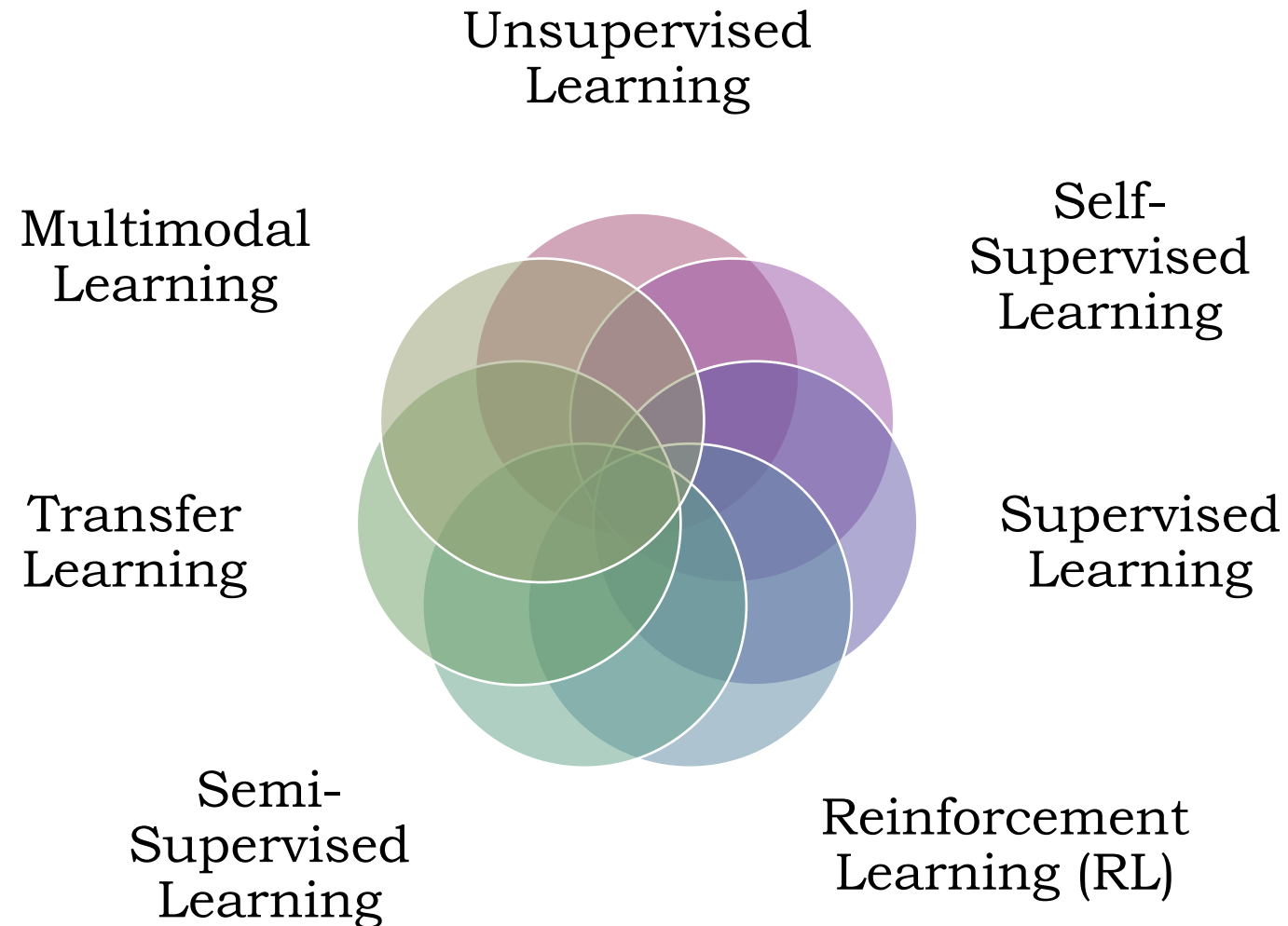
- **Audio generation**

- Tools like OpenAI's MuseNet can compose original music in various genres, helping musicians create new pieces, soundtracks, and soundscapes for film.

- **Video creation**

- Models like Synthesia can create synthetic videos featuring realistic avatars that deliver presentations or narrate content, streamlining educational and marketing efforts.

Types of Learning in Generative AI

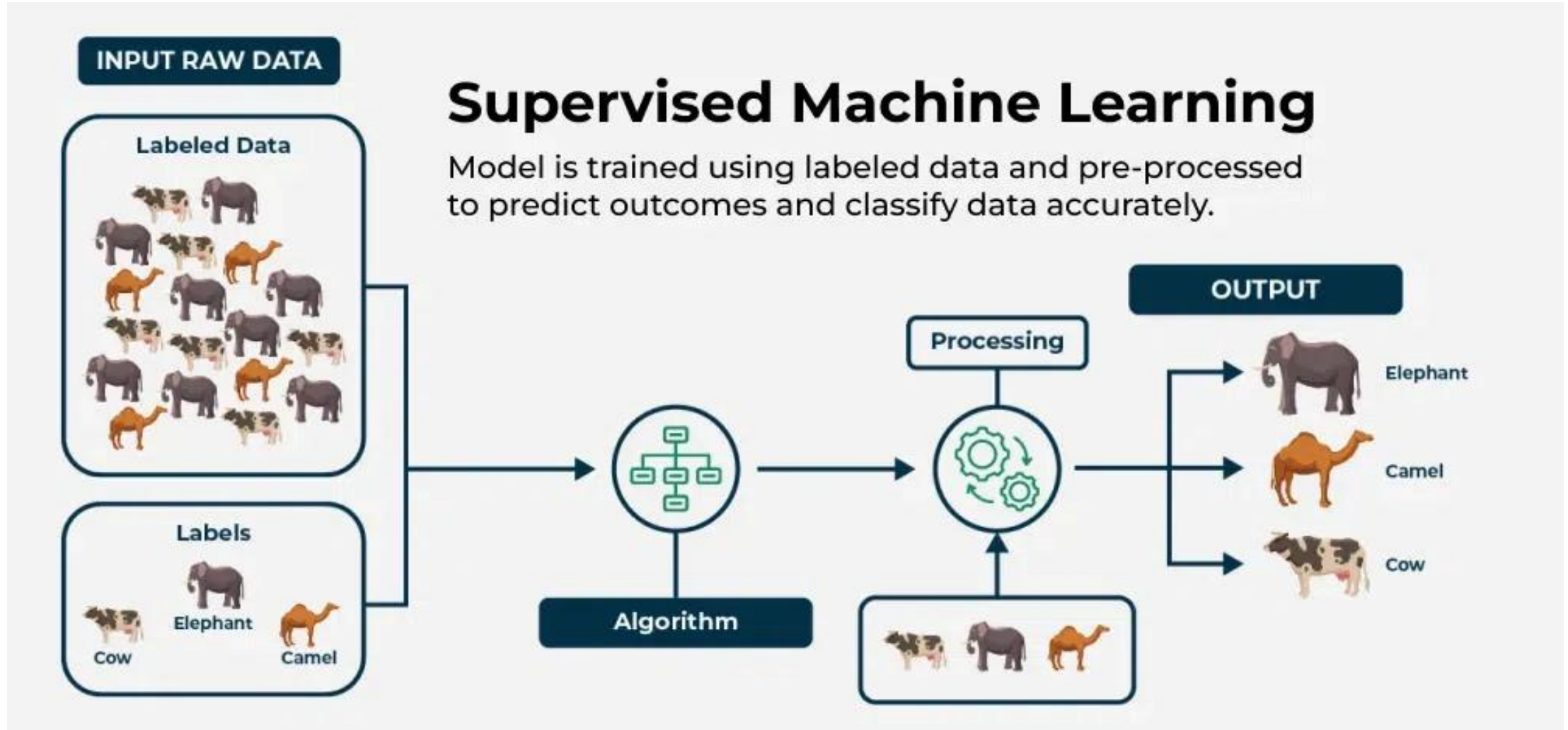


Supervised Learning

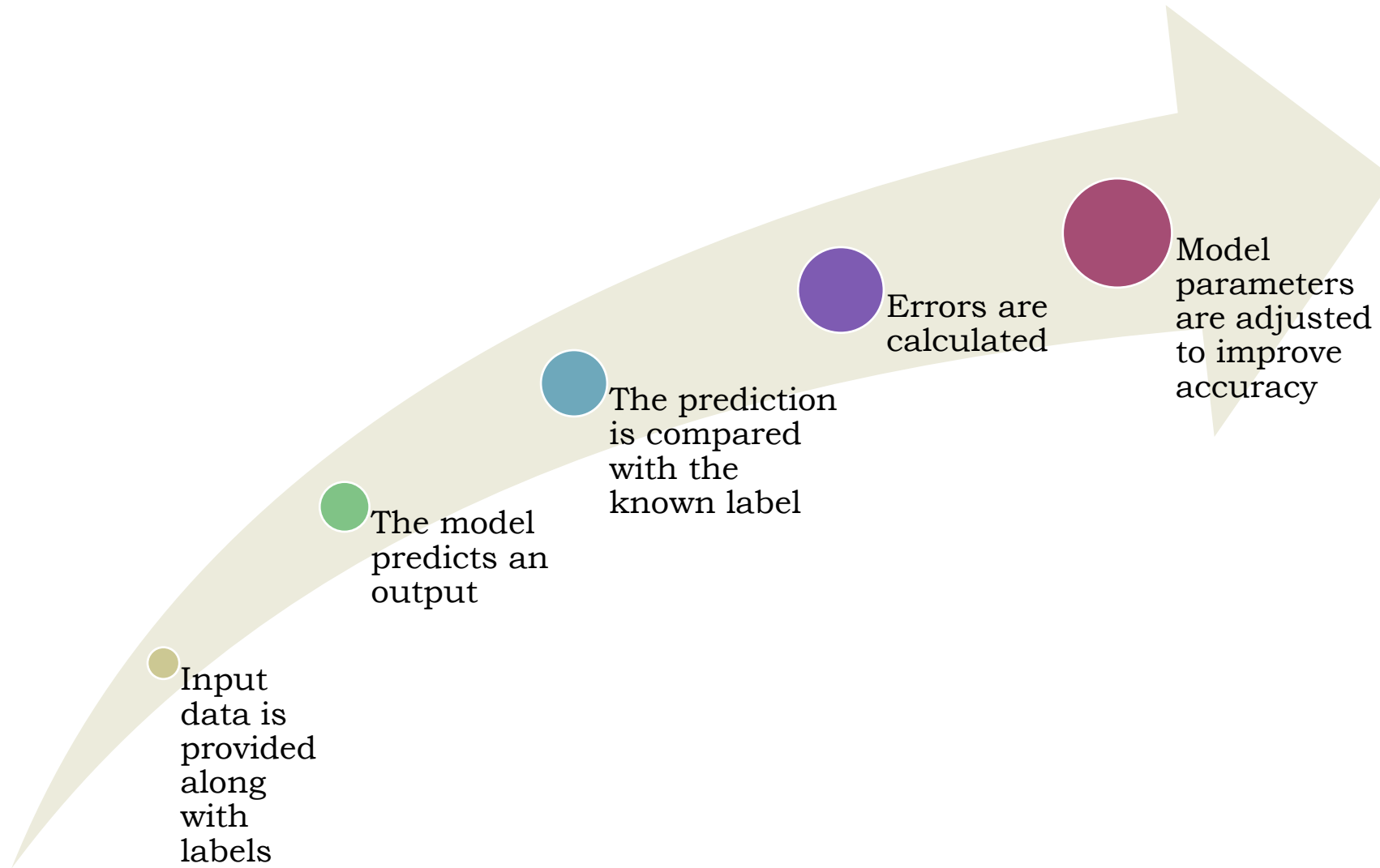
- Supervised Learning is a machine learning approach where a model is trained using labeled data, meaning each input is paired with a known and correct output.
- The model makes predictions and compares them with the true outputs, adjusting itself to reduce errors and improve accuracy over time.
- The goal is to make accurate predictions on new, unseen data.
- Types of Supervised Learning:
 - **Classification** – where the out is categorical variable.
 - **Regression** – where the output is a continuous variable.



Supervised Learning



How Supervised Learning Works?



Examples of Supervised Learning

Email spam
detection

Image
classification

Fraud
detection

Credit
scoring

Speech
recognition

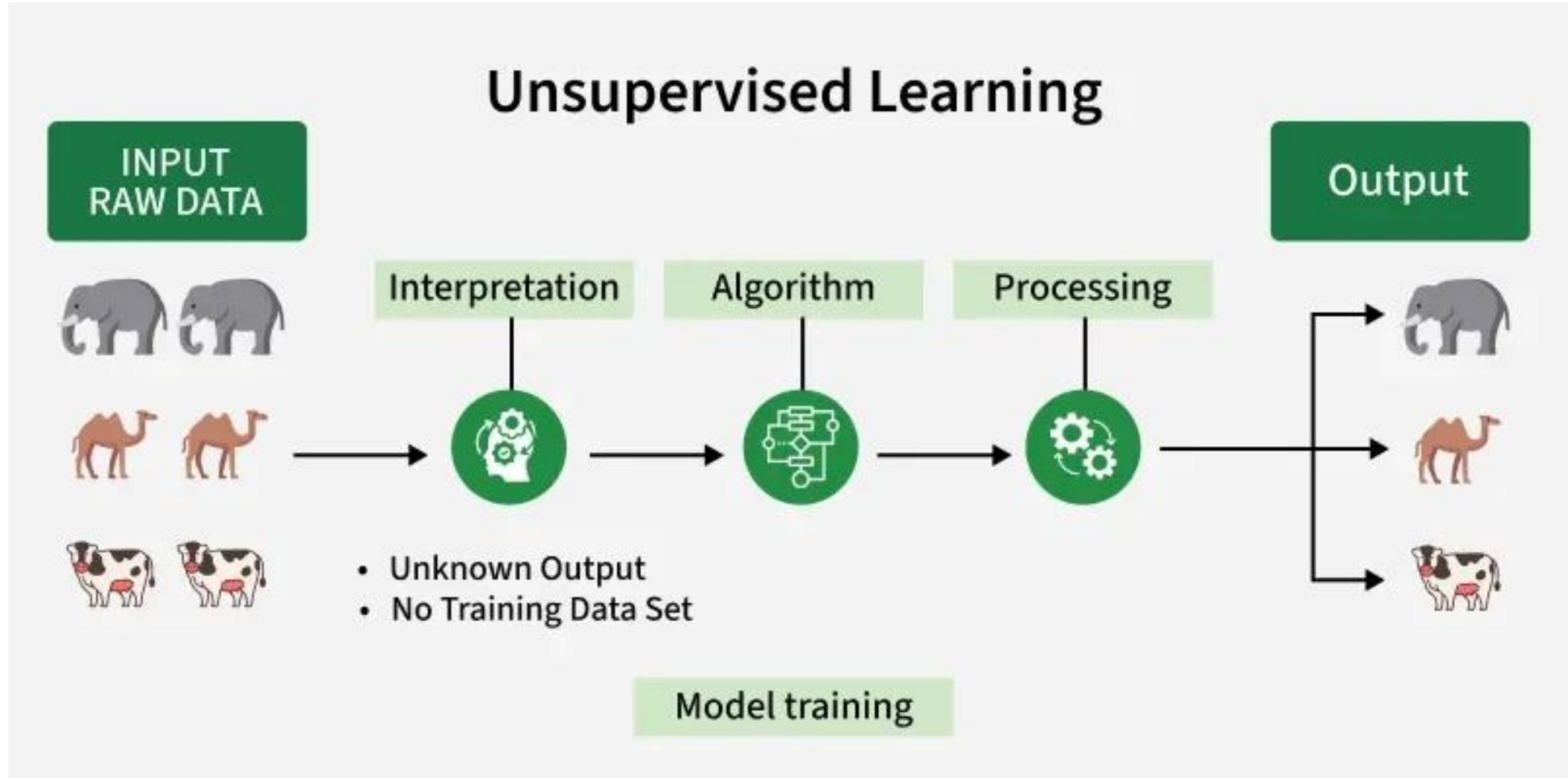


Unsupervised Learning

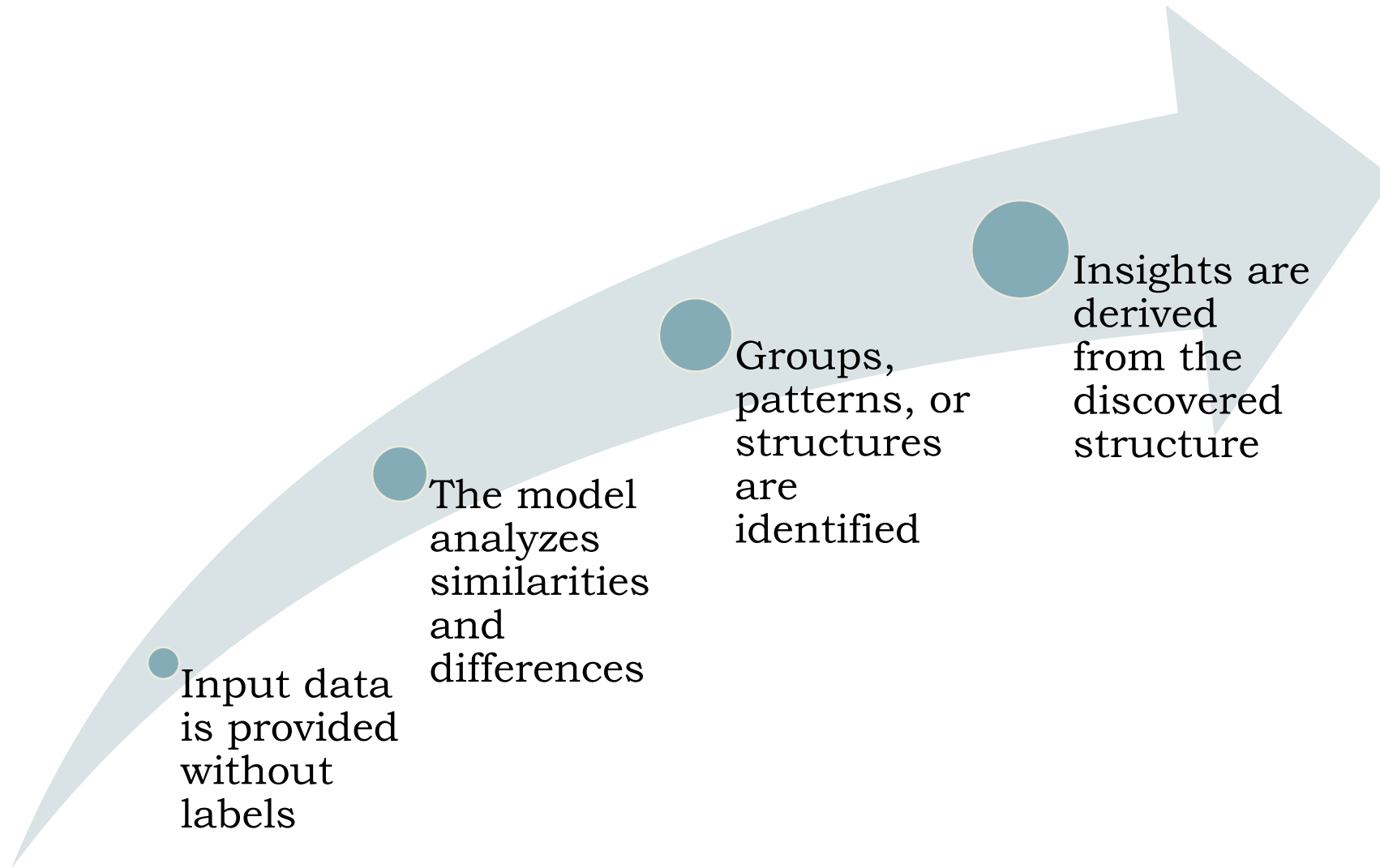
- Unsupervised Learning is a machine learning approach where the model is trained on unlabeled data, and it independently discovers patterns, structures, or relationships.
- Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.
- It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.
- Helps identify hidden patterns in data
- Useful for grouping, compression and anomaly detection



Unsupervised Learning



How Unsupervised Learning Works?



Examples of Unsupervised Learning

Customer
segmentation

Market
basket
analysis

Topic
modeling

Network
anomaly
detection



Discriminative AI

- Discriminative AI refers to a class of artificial intelligence models that are designed to classify, predict, or make decisions by learning the direct relationship between input data and output labels.
- Discriminative AI learns decision boundaries between classes to accurately classify or predict outcomes.
- Common Discriminative Models:
 - Logistic Regression
 - Support Vector Machines (SVM)
 - Decision Trees
 - Random Forest
 - Discriminative Neural Networks



Examples of Discriminative AI

- Email spam detection
- Fraud detection systems
- Face recognition
- Credit risk scoring
- Malware classification



Common Discriminative Models

Logistic
Regression

Support Vector
Machines
(SVM)

Decision Trees

Random Forest

Discriminative
Neural
Networks



Advantages & Disadvantages

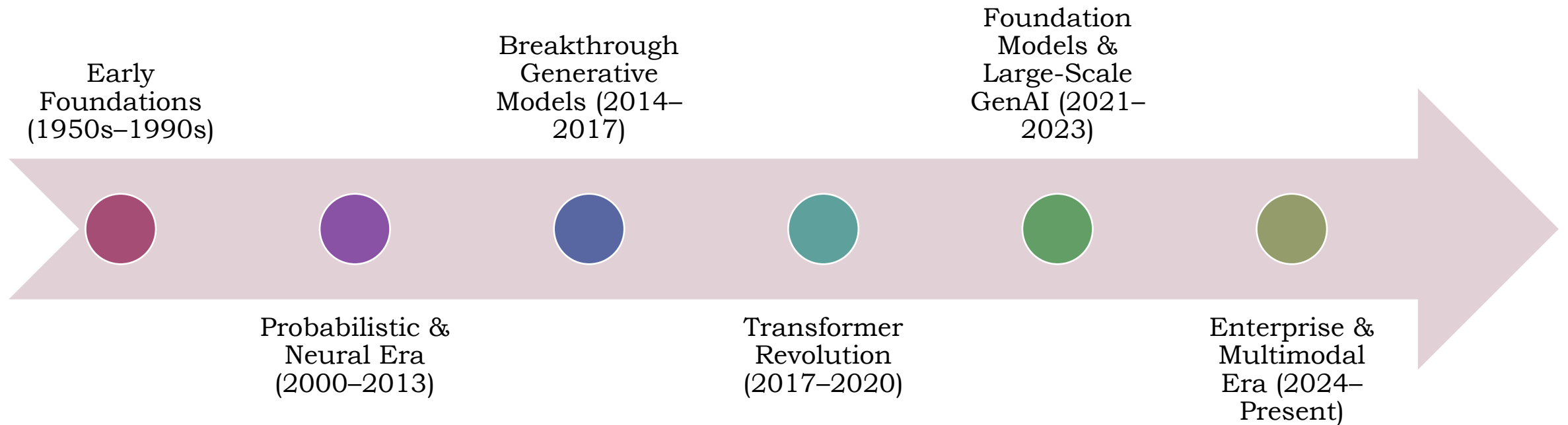
- **Advantages**

- Efficient and fast
- Easy to evaluate and validate
- Well-suited for real-time decision systems

- **Disadvantages**

- Cannot create new content
- Limited understanding of data generation process
- Less flexible compared to generative models

Brief History of Generative AI



Brief History of Generative AI

▪ **Early Foundations (1950s–1990s)**

- 1950: Alan Turing proposes the Turing Test, laying the philosophical foundation for machine intelligence.
- 1950s–1980s: Rule-based AI and expert systems dominate; limited generative capability.
- 1980s–1990s: Statistical models (e.g., Hidden Markov Models) used for speech and text generation in constrained domains.
- Key Limitation: Systems relied on handcrafted rules and had no true creativity or learning at scale.

▪ **Probabilistic & Neural Era (2000–2013)**

- Introduction of probabilistic models for generation.
- 2006: Geoffrey Hinton popularizes deep learning.
- Early neural language models begin generating simple text sequences.
- Key Advancement: Models start learning patterns directly from data rather than rules.



Brief History of Generative AI

▪ **Breakthrough Generative Models (2014–2017)**

- 2014: Generative Adversarial Networks (GANs) introduced by Ian Goodfellow.
- Variational Autoencoders (VAEs) enable structured data generation.
- Significant progress in image and audio generation.
- Impact: AI can now generate realistic images, faces, and signals.

▪ **Transformer Revolution (2017–2020)**

- 2017: Google introduces the Transformer architecture (“Attention Is All You Need”).
- Models scale effectively using attention mechanisms.
- 2018–2020: Large language models (GPT, BERT variants) emerge.
- Impact: High-quality text generation, translation, summarization, and context awareness.

Brief History of Generative AI

- **Foundation Models & Large-Scale GenAI (2021–2023)**

- Emergence of Foundation Models trained on massive datasets.
- Text-to-image, text-to-code, and multimodal generation become mainstream.
- Public adoption accelerates through conversational AI systems.
- Key Shift: Generative AI becomes usable by non-technical users.

- **Enterprise & Multimodal Era (2024–Present)**

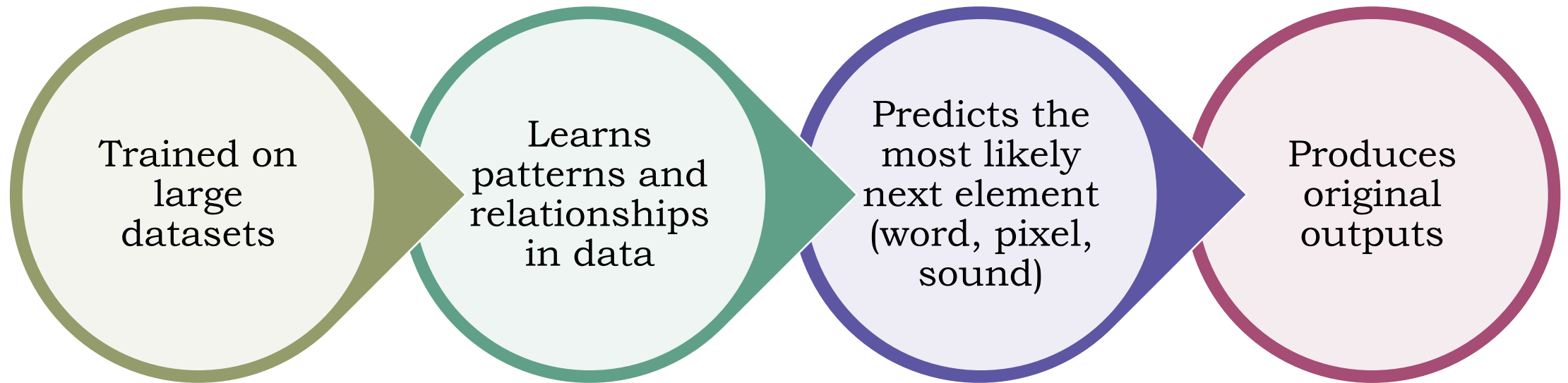
- Integration of GenAI into enterprise systems, security, development, and operations.
- Multimodal models handle text, images, audio, video, and code together.
- Focus on governance, safety, compliance, and responsible AI.
- Current State: Generative AI is a strategic business capability, not just a research tool.

Generative AI

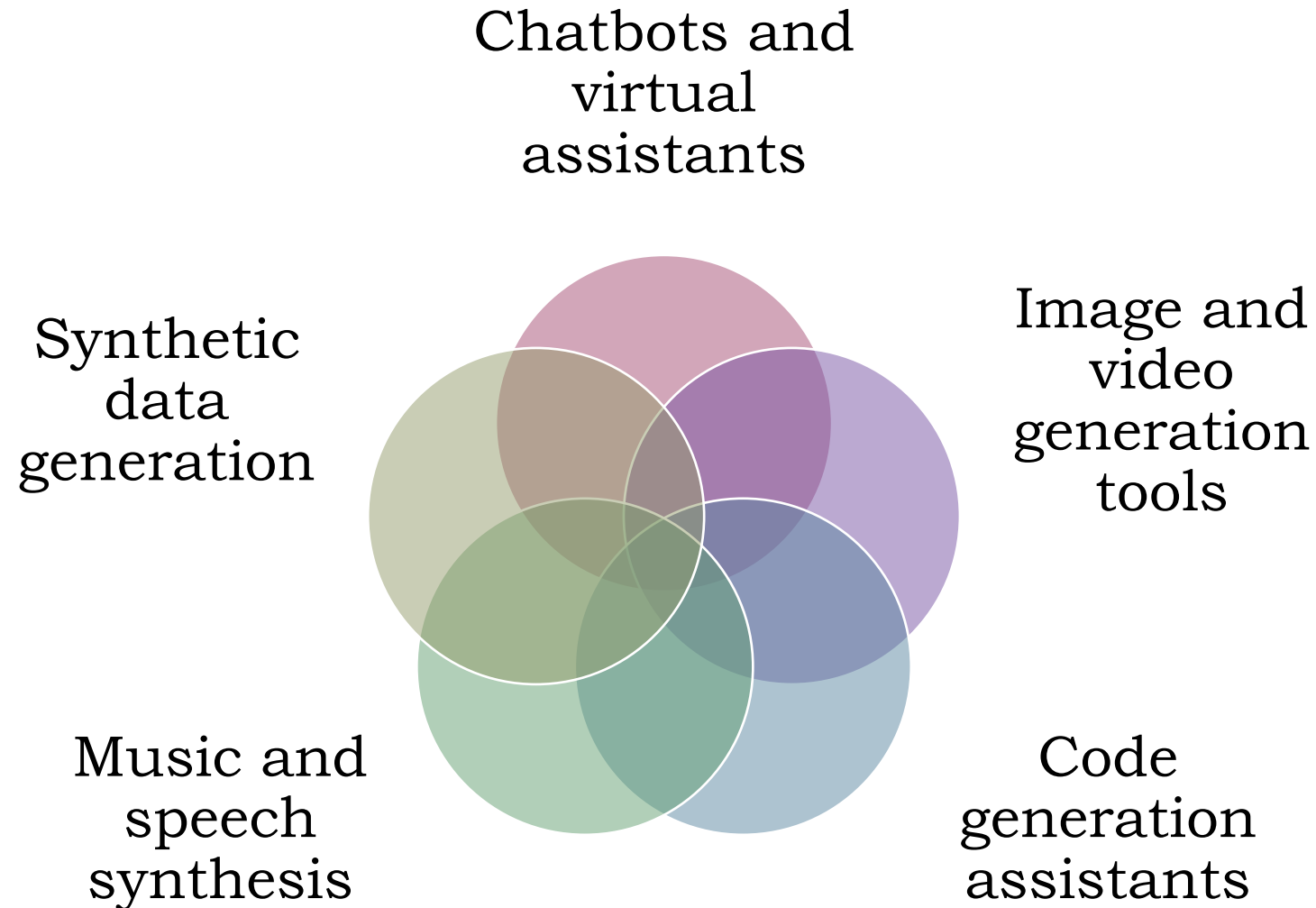
- Generative AI refers to models that learn the underlying data distribution and can generate new, original data similar to the training data.
- Generative AI refers to a class of AI systems that are designed to create new, original content such as text, images, audio, video, code, or synthetic data by learning patterns and structures from large datasets.
- Key Characteristics
 - Generates new content, not just predictions
 - Learns the underlying data distribution
 - Produces probabilistic and context-aware outputs
 - Adapts outputs based on prompts or inputs



How Generative AI Works?



Examples of Generative AI



Discriminative AI vs Generative AI

Aspect	Discriminative AI	Generative AI
Definition	Learns decision boundaries	Learns data distribution
Objective	Predict labels/outcomes	Generate new data
Core Function	Classification/Regression	Content Generation
Output	Labels, scores	Text, images, audio
Data Type	Labeled data	Unlabeled / large-scale
Learning Type	Supervised	Self/Unsupervised
Focus	Prediction accuracy	Content quality
Complexity	Lower	Very high
Training Cost	Low to moderate	Very high
Model Examples	SVM, Random Forest	GANs, Transformers
Enterprise Use	Fraud detection	Chatbots, media creation
Explainability	High	Low
Security Risk	Low	High
Real Examples	Spam filter	ChatGPT, DALL·E
System Role	Decision maker	Content creator



A Peek into Generative AI Models

- **Language Models (LLMs)**

- Generate and understand human language

- **Generative Adversarial Networks (GANs)**

- Generate highly realistic images and media

- **Variational Autoencoders (VAEs)**

- Generate structured and controllable outputs

- **Diffusion Models**

- High-quality image, audio, and video generation

- **Multimodal Models**

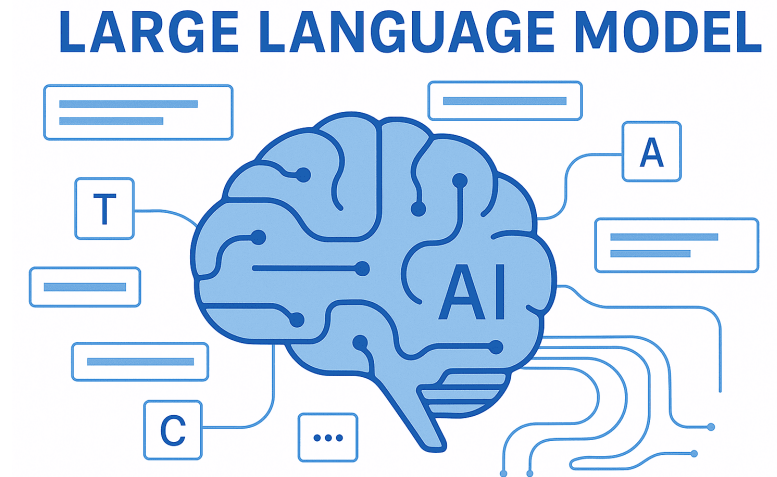
- Handle multiple data types simultaneously

- **Code Generative Models**

- Generate and understand programming code

Language Models (LLMs)

- Language Models, commonly referred to as Large Language Models (LLMs), are a class of artificial intelligence systems designed to understand, generate, and manipulate human language.
- A Language Model is an AI model that assigns probabilities to sequences of words or tokens, enabling it to predict the next token in a given context.
- When scaled to billions or trillions of parameters and trained on massive datasets, these models are called Large Language Models (LLMs).
- Examples of LLMs
 - GPT (Generative Pre-trained Transformer)
 - BERT (Bidirectional Encoder Representations from Transformers)
 - LLaMA, PaLM

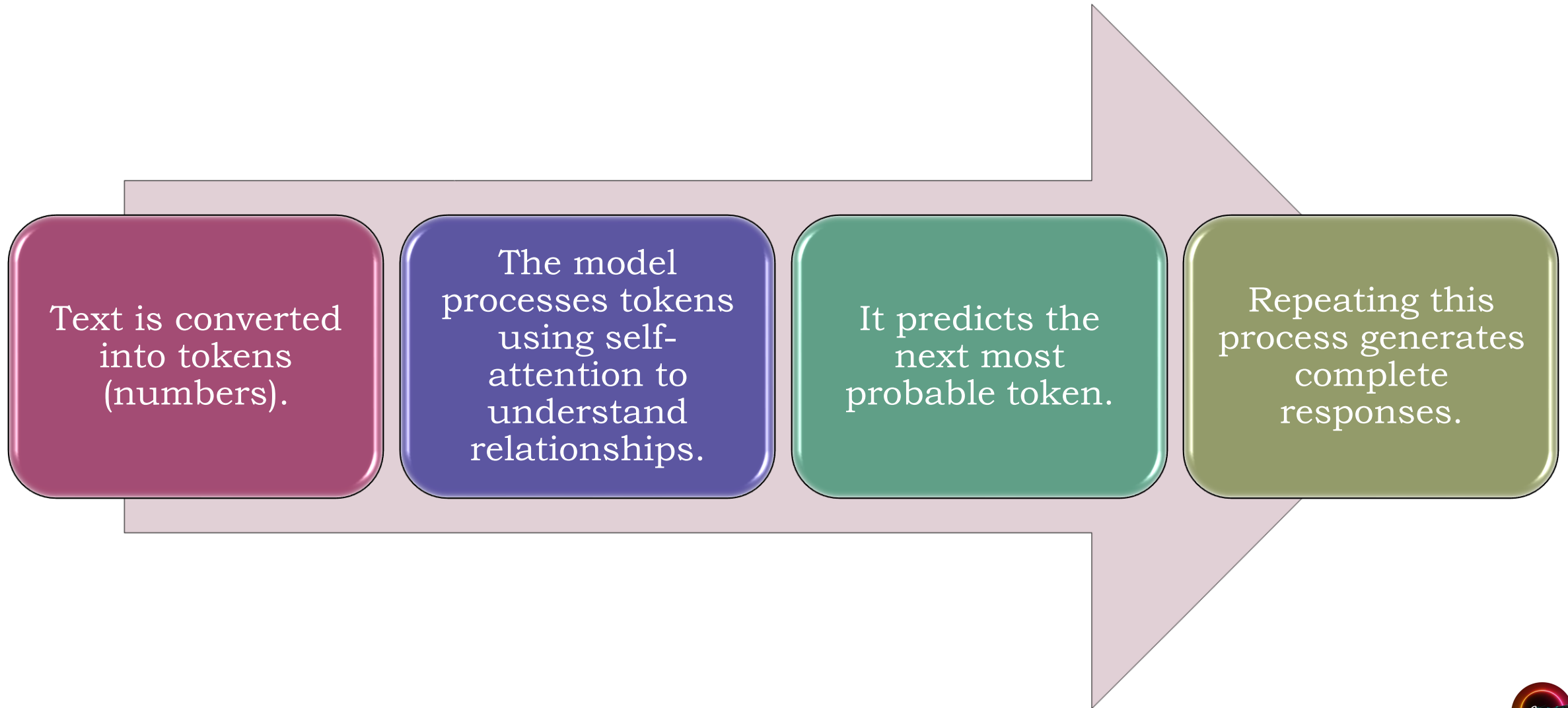


Key Characteristics of LLMs

- **Context Awareness:** Understands meaning based on surrounding words, sentences, or paragraphs.
- **Generative Capability:** Produces coherent and contextually relevant text.
- **Pre-trained on Large Corpora:** Trained on books, articles, code, and web-scale text.
- **Transformer-Based Architecture:** Typically built using the Transformer neural network with attention mechanisms.
- **Fine-Tunable:** Can be adapted for specific tasks or domains (e.g., healthcare, cybersecurity, enterprise IT).



How LLMs Work



Common Tasks Performed by LLMs

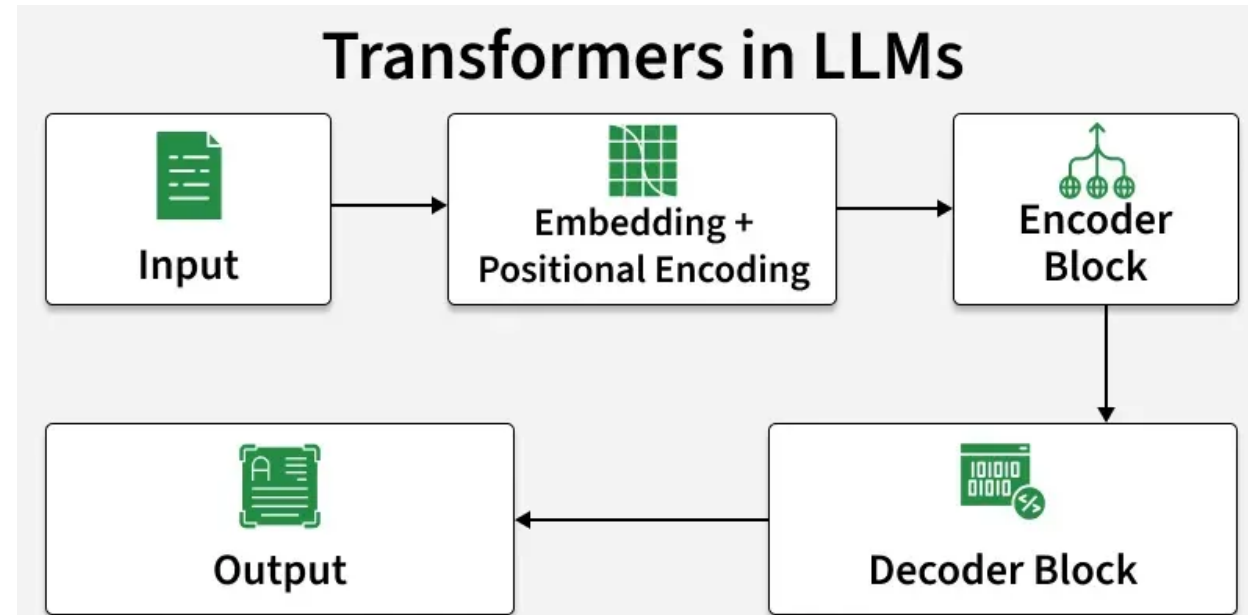
- Text generation and summarization
- Question answering and chatbots
- Translation and sentiment analysis
- Code generation and explanation
- Document analysis and knowledge extraction

Working of LLM

LLMs are primarily based on the Transformer architecture which enables them to learn long-range dependencies and contextual meaning in text.

At a high level:

- **Input Embeddings:** Converting text into numerical vectors.
- **Positional Encoding:** Adding sequence/order information.
- **Self-Attention:** Understanding relationships between words in context.
- **Feed-Forward Layers:** Capturing complex patterns.
- **Decoding:** Generating responses step-by-step.
- **Multi-Head Attention:** Parallel reasoning over multiple relationships.



Popular LLMs

Some of the most widely used LLMs include:

- **GPT-4 and GPT-4o (OpenAI):** Advanced multimodal reasoning and dialogue capabilities.
- **Gemini 1.5 (Google DeepMind):** Long-context reasoning, capable of handling 1M+ tokens.
- **Claude 3 (Anthropic):** Safety-focused, strong at reasoning and summarization.
- **LLaMA 3 (Meta):** Open-weight model, popular in research and startups.
- **Mistral 7B / Mixtral (Mistral AI):** Efficient open-source alternatives for developers.
- **BERT and RoBERTa (Google/Facebook):** Strong embedding models for NLP tasks.
- **mBERT and XLM-R:** Early multilingual LLMs.
- **BLOOM:** Large open-source multilingual model, collaboratively developed.



Use Cases

- **Code Generation:** LLMs can generate accurate code based on user instructions for specific tasks.
- **Debugging and Documentation:** They assist in identifying code errors, suggesting fixes and even automating project documentation.
- **Question Answering:** Users can ask both casual and complex questions, receiving detailed, context-aware responses.
- **Language Translation and Correction:** LLMs can translate text between over 50 languages and correct grammatical errors.
- **Prompt-Based Versatility:** By crafting creative prompts, users can unlock endless possibilities, as LLMs excel in one-shot and zero-shot learning scenarios.



Advantages of LLMs

- **Zero-Shot and Few-Shot Learning:** Can perform new tasks without explicit retraining.
- **Scalable Knowledge:** Efficiently process and understand vast text corpora.
- **Fine-Tuning Flexibility:** Adaptable to specific industries and datasets.
- **Automation of Language Tasks:** Frees human effort from repetitive or time-consuming tasks.
- **Versatility:** Effective across multiple domains—healthcare, education, business and research.



High-Level Model Comparison

Model Type	Best At	Output Quality	Speed	Stability
LLMs	Text & code	High	Fast	High
GANs	Realistic images	Very High	Fast	Medium
VAEs	Structured generation	Medium	Fast	High
Diffusion	Artistic realism	Very High	Slow	Very High
Multimodal	Cross-domain tasks	Very High	Medium	High

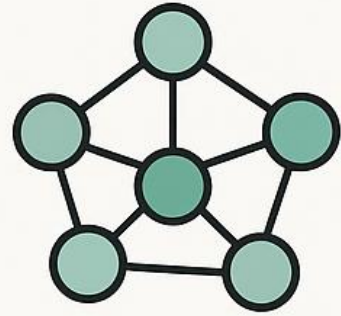
ChatGPT

- ChatGPT is a conversational Artificial Intelligence system developed using Generative AI and Large Language Models (LLMs).
- It is designed to understand, generate, and interact using natural human language.
- Core Capabilities
 - Natural language conversations
 - Question answering and explanations
 - Code generation and debugging
 - Summarization and translation
 - Reasoning and problem-solving
 - Content creation (emails, reports, training material)



ChatGPT

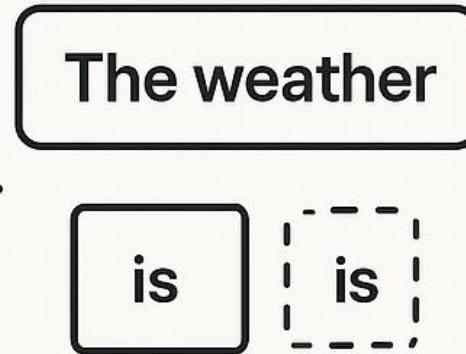
How ChatGPT Works ?



Built on
Transformer-
based neural
networks



Trained on
large-scale
text data
using self-
supervised
learning



Predicts the
next word
(token) based
on context



Fine-tuned using
human feedback
to improve
safety and
usefulness

Key Features & Benefits of ChatGPT

Key Features:

- Context-aware responses
- Multi-domain knowledge (technology, business, education)
- Supports structured and unstructured queries
- Scales from simple Q&A to complex analysis

Benefits:

- Improves productivity and efficiency
- Reduces time spent on repetitive tasks
- Enables non-technical users to leverage AI
- Enhances decision support

ChatGPT vs Traditional Chatbots

Aspect	Traditional Chatbots	ChatGPT
Intelligence	Rule-based	Generative AI-based
Flexibility	Low	High
Learning Capability	Static	Dynamic (pre-trained)
Conversation Quality	Scripted	Natural & contextual

How does Model understand Text?

1. Text Is Converted into Tokens
2. Tokens Are Mapped to Numbers
3. Embeddings Capture Meaning
4. Context Is Learned Using Attention
5. Neural Layers Build Understanding
6. Probability-Based Language Modeling
7. Fine-Tuning with Human Feedback

1. Text Is Converted into Tokens

- Raw text is first broken into tokens (words, subwords, or characters).
- Example:
 - “Artificial Intelligence” → ["Artificial", "Intelli", "gence"]
- **Why this matters:**
- Tokens standardize language and reduce vocabulary size.



2. Tokens Are Mapped to Numbers

- Each token is converted into a numerical ID.
- AI models operate only on numbers, not text.
- Example:
 - "Artificial" → 10245



3. Embeddings Capture Meaning

- Token IDs are transformed into vectors (embeddings).
- Embeddings represent semantic meaning and relationships.
- Similar words have similar vectors.
- Example:
 - “king” and “queen” are closer in vector space than “king” and “table”.

4. Context Is Learned Using Attention

- Transformer models use self-attention mechanisms.
- Each word attends to other words to understand context and relationships.
- Example:
 - In the sentence “He deposited money in the bank”,
 - attention helps associate bank with finance, not a river.



5. Neural Layers Build Understanding

- Multiple layers refine embeddings progressively.
- Lower layers capture syntax
- Higher layers capture semantics and intent

6. Probability-Based Language Modeling

- The model learns to predict the next token based on previous tokens.
- This statistical learning enables comprehension-like behavior.
- Example:
 - “I am going to the” → highest probability word = “office” or “market”



7. Fine-Tuning with Human Feedback

- Models are refined using human-labeled data.
- Improves correctness, safety, and usefulness.







That's all Folks!